Astronomy - Graduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.
   (a) It has a fusion crust
   (b) It contains solidified spherical droplets
   (c) It is highly processed
   (d) It has different elemental composition than earth
2. What would weigh the most on the moon?
   (a) A kilogram of feathers
   (b) Five pounds of bricks as measured on Earth
   (c) Five kilograms of feathers
   (d) A kilogram of bricks
3. Which of the following most likely explains why Venus does not have a strong magnetic field?
   (a) Its rotation is too slow.
   (b) It has too thick an atmosphere.
   (c) It is too large.
   (d) It does not have a metallic core.
4. Which living organisms most resemble the common ancestor of all life according to genetic testing?
   (a) viruses
   (b) bacteria such as E. coli
   (c) organisms living deep in the oceans around seafloor volcanic vents and in hot springs
   (d) plankton that use sunlight as an energy source through photosynthesis
5. Which is the least likely cause of death?
   (a) Being hit in the head by a bullet.
   (b) Being hit by a small meteorite.
   (c) Starvation during global winter caused by a major impact.
   (d) Driving while intoxicated without wearing seatbelts.

True / False (10 marks)

Answer True or False

6. True or False: A lunar day lasts about 24 hours, similar to a day on Earth.
7. True or False: Callisto is the closest of Jupiter's four largest moons, with a smaller distance than Io.
8. True or False: Kirkwood gaps are found where asteroids would orbit with a period twice that of Mars.
9. True or False: When observing a star's spectrum through cool hydrogen gas, the blackbody spectrum displays absorption lines from hydrogen.
10. True or False: The orbital speed of the International Space Station (ISS) is approximately 28000 km/h.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of rapid rotation in the formation of the equatorial bulge observed in jovian planets, analyzing the underlying physical forces and implications for their structure.
12. Discuss how the angular resolution of the human eye affects the observation of closely spaced stars, and explain the implications for astronomical observations.

End of Paper